Part II: Experiment Details

1. Setup and Environment

- **Platform:** Kaggle Notebooks
- **Hardware:** 2x T4 GPUs (Parallel)
- **Framework:** PyTorch with Mixed Precision (AMP)
- **Time Allocated:** ~12 Hours Total (10 hours for training: 5 hours each for BERT and GPT)

2. Hyperparameters

Parameter	BERT Encoder	GPT Decoder
Vocab Size	16,001	16,000
Hidden Dim	768	768
Layers	12	12
Attention Heads	12 (4 KV heads for GQA)	12 (4 KV heads for GQA)
FFN Dim	4096	4096
Max Sequence	256	256
Batch Size	32	64
Learning Rate	1e-4	1e-4
Positional Encoding	RoPE	RoPE
Normalization	RMSNorm	RMSNorm

3. BERT Pretraining Logs (Masked Language Modeling)

- **Dataset:** `train.hi` and `val.hi`
- **Total Training Time:** ~5 Hours

- **Max Epoch Reached:** Epoch 6 (stopped early due to time constraints)
- **Checkpoint Strategy:** Saved mid-epoch every 2000 steps, and comprehensive checkpoint at the end of each epoch.

Epoch	Train Loss (Final Step)	Validation Loss	Validation PPL
0	4.840	4.731	113.41
1	4.241	4.230	68.74
2	3.980	3.978	53.41
3	3.466	3.826	45.88
4	3.273	2.534	12.60
5	2.704	*Incomplete*	*Incomplete*

4. GPT Pretraining Logs (Causal Language Modeling)

- **Dataset:** `train.mr` and `val.mr`
- **Total Training Time:** ~5 Hours
- **Max Epoch Reached:** Epoch 9 (crashed due to instability)
- **Checkpoint Utilized:** Epoch 8

Epoch	Train Loss (Final Step)	Validation Loss	Validation PPL
6	2.268	3.597	36.50
7	2.202	3.598	36.55
8	2.266	3.538	34.41

5. Bridged Translation (Zero-Shot Soft-Prompting)

- **Technique:** Concatenating BERT output embeddings with GPT token embeddings.
- **Trained Parameters during NMT:** None (Randomly initialized projection layer `768 -> 768`).

- **Inference Time:** ~2 hours (including metric evaluation and script compilation).
- **Final Metrics:** Corpus BLEU = 0.00, CHRF++ = 10.44.